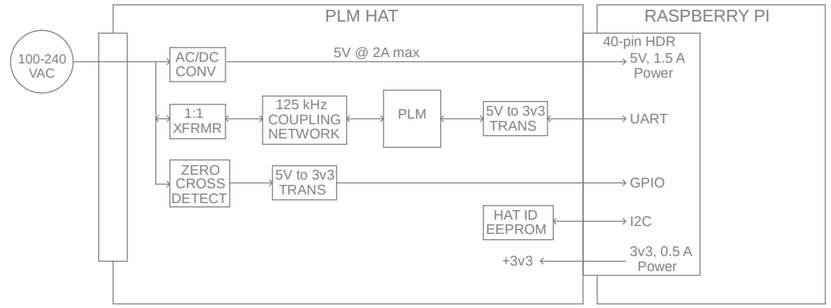
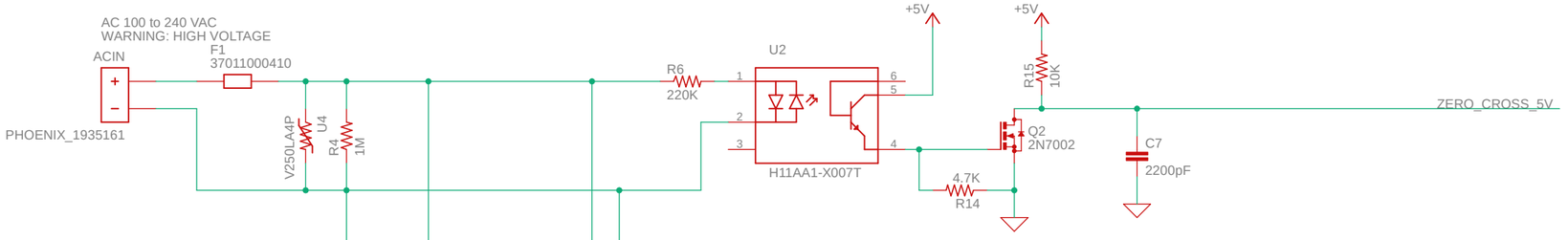


BLOCK DIAGRAM

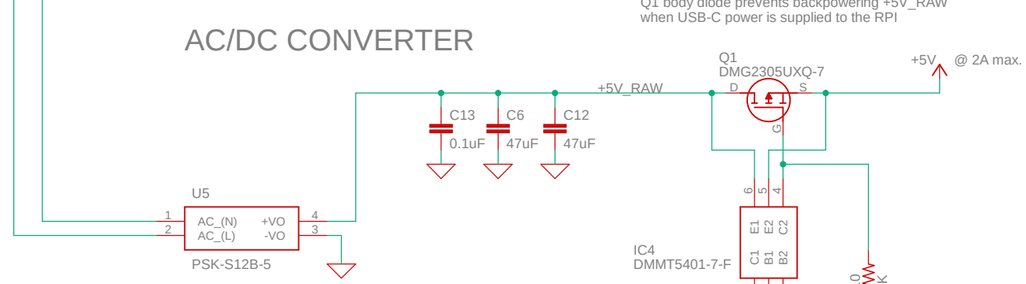
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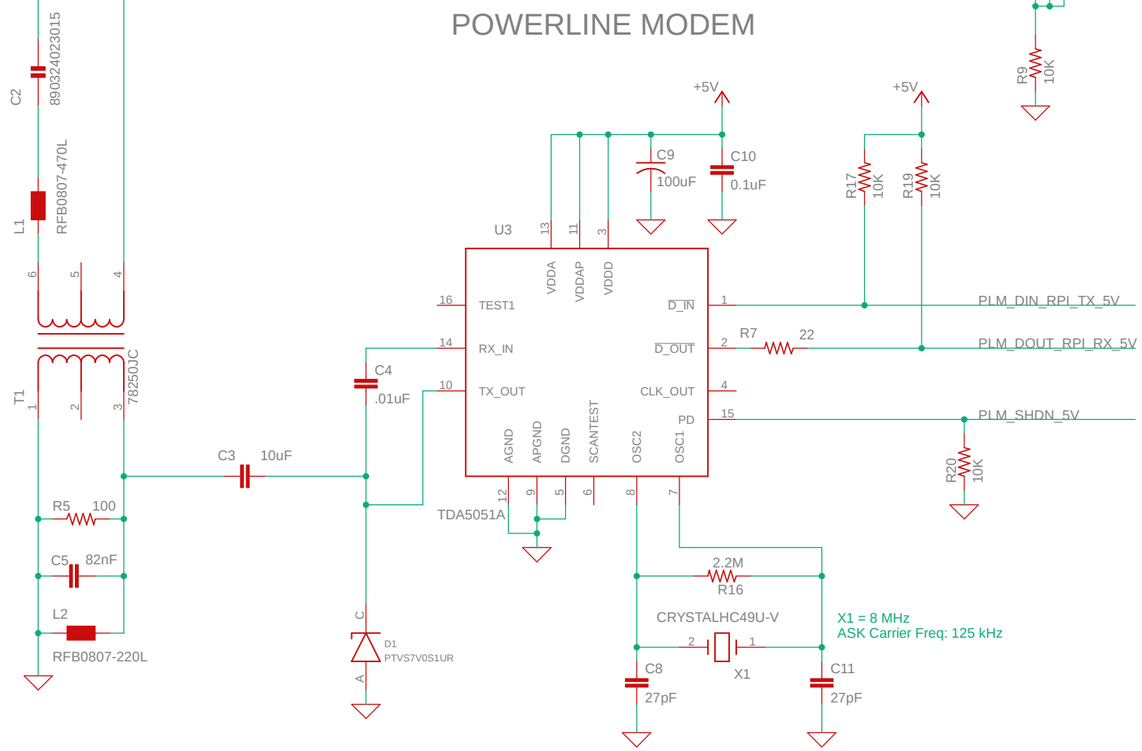
ZERO CROSS DETECTOR



AC/DC CONVERTER



POWERLINE MODEM

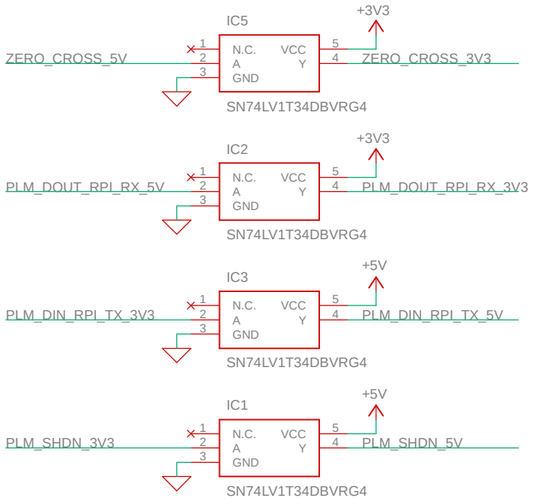


ZC DETECTOR, POWER, PLM

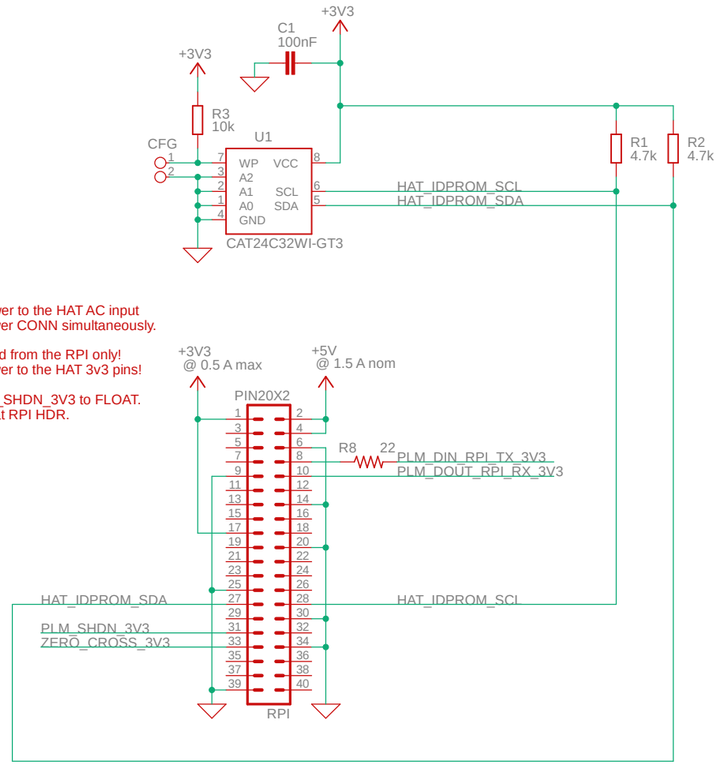
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RPI to PLM DIO LEVEL TRANSLATION



RPI HDR & ID EEPROM



NOTE:
-DO NOT supply power to the HAT AC input and RPI USB-C power CONN simultaneously.

-3v3 power is sourced from the RPI only! DO NOT supply power to the HAT 3v3 pins!

-DO NOT allow PLM_SHDN_3V3 to FLOAT. Pin must be driven at RPI HDR.

RASPBERRY PI 40-PIN GPIO HEADER PINOUT

PIN #	RPI SIGNAL	HAT SIGNAL
1	3v3 Power	3v3 Power
2	5V Power	5V Power
3	GPIO2 (SDA)	N.C.
4	5V Power	5V Power
5	GPIO3 (SCL)	N.C.
6	GND	GND
7	GPIO4 (GPCLK0)	N.C.
8	GPIO14 (TXD)	PLM_DIN_RPI_TX_3V3
9	GND	GND
10	GPIO15 (RXD)	PLM_DOUT_RPI_RX_3V3
11	GPIO17	N.C.
12	GPIO18 (PCM_CLK)	N.C.
13	GPIO27	N.C.
14	GND	GND
15	GPIO22	N.C.
16	GPIO23	N.C.
17	3v3 Power	3v3 Power
18	GPIO24	N.C.
19	GPIO10 (MOSI)	N.C.
20	GND	GND
21	GPIO9 (MISO)	N.C.
22	GPIO25	N.C.
23	GPIO11 (SCLK)	N.C.
24	GPIO8 (CE0)	N.C.
25	GND	GND
26	GPIO7 (CE1)	N.C.
27	GPIO0 (ID_SD)	HAT_IDPROM_SDA
28	GPIO1 (ID_SC)	HAT_IDPROM_SCL
29	GPIO5	N.C.
30	GND	GND
31	GPIO6	PLM_SHDN_3V3
32	GPIO12 (PWM0)	N.C.
33	GPIO13 (PWM1)	ZERO_CROSS_3V3
34	GND	GND
35	GPIO19 (PCM_FS)	N.C.
36	GPIO16	N.C.
37	GPIO26	N.C.
38	GPIO20 (PCM_DIN)	N.C.
39	GND	GND
40	GPIO21 (PCM_DOUT)	N.C.

HAT RPI INTERFACE & EEPROM

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